

2 Atoms, molecules and stoichiometry

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Use $n = m/M$, $n = cV$ with V in dm^3 , and gas molar volume only under specified conditions. Balance equations before mole ratios. A limiting reagent runs out first; compare available amount divided by stoichiometric coefficient. Percentage yield compares actual and theoretical product amounts.

Worked example: follow the reasoning

$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$: 0.10 mol Mg needs 0.050 mol O_2 and forms 0.10 mol MgO if fully reacted.

Quick check · multiple choice

A Which reagent limits when 1 mol N_2 reacts with 1 mol H_2 to make NH_3 ? [1]

- A Nitrogen
- B Hydrogen
- C Both equal
- D Neither

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 0.20 mol N_2 reacts with 0.45 mol H_2 : $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. Find limiting reagent and theoretical NH_3 amount. [3]

Final answer: _____

Answers and explanation

Quick check: B. Hydrogen

Question 1 [3 marks]

0.20 mol N₂ needs 0.60 mol H₂, so H₂ limits [1]. NH₃ = (2/3)(0.45) = 0.30 mol [2].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.